Agri-Brain — Process Flow & Use-Case Diagram

AMD Slingshot 2026 | AI-Orchestrated Rural Grid | System Interaction Overview

① HIGH-LEVEL SYSTEM PROCESS FLOW

Field Sensors
ESP32 reads Moisture · pH · Rain · PSI

→

MQTT Broker
HiveMQ TLS / Mosquitto local relay

→

AMD AI Gateway
Python orchestrator runs all AI brains

→

AI Decision Engines
Acoustic · Vision · NLP · Soil · Gemini

→

MQTT Command
control/grid/ON · OFF · alerts

→

Valve / Field Action
ESP32 activates grid valves

② DETAILED SWIMLANE PROCESS FLOW

ACTOR	PROCESS STEPS
FARMER	**Voice / Manual** (Voice cmd or web tap) → **Upload Leaf Photo** (Via web dashboard) → **View Alerts** (Dashboard / mobile) → **Review Ledger** (Farm activity log)
ESP32 FIELD	**Read Sensors** (DHT22 · pH · PSI) → **Publish Telemetry** (MQTT every 5s) → **Receive Command** (control/grid/ON-OFF) → **Actuate Valve** (Solenoid open/close) → **Safety Timeout** (2-hr hard cutoff)
MQTT BROKER	**telemetry/region/*** (Sensor data in) ↔ **voice_command** (Voice/text input) ↔ **control/grid/*** (Commands out) ↔ **ledger/update** (Activity log push) ↔ **ai/soil/alert** (Health alerts push)
AMD GATEWAY	**Parse Telemetry** (verify_mqtt.py) → **Soil Brain** (pH+moisture analysis) → **NLP Brain** (Parse voice intent) → **Irrigation Queue** (Sequential dispatch) → **Acoustic Brain** (PSI guard / E-stop)
	Receive Image → **vision_brain.py** → **Disease / Pest Alert** → **Gemini Advisor** → **Farmer Notified**

VISION + AI

From web upload → Local AMD inference → Publish to dashboard → Treatment advice → Web + Mobile alert

③ USE-CASE DIAGRAM — AGRI-BRAIN SYSTEM

ACTORS

- Farmer
- ESP32 / Sensors
- AMD AI Gateway
- Gemini API

«SYSTEM» AGRI-BRAIN PLATFORM

UC-01: Monitor Soil Health

Actor: ESP32, AMD Gateway

- ESP32 reads moisture + pH
- Publishes to MQTT telemetry
- Soil Brain analyses per-grid
- Alert if below threshold

UC-02: Auto Irrigation

Actor: AMD Gateway, ESP32

- Moisture < 50% detected
- Grid added to queue
- Sequential valve opens
- Timer auto-closes valve

UC-03: Manual Irrigation

Actor: Farmer, AMD Gateway

- Farmer taps web/mobile
- Selects region + duration
- MQTT command dispatched
- ESP32 opens valve

UC-04: Voice Command

Actor: Farmer, AMD Gateway

- Farmer speaks in Kannada/Hindi
- NLP Brain parses intent
- Irrigation / logging action
- Voice feedback returned

UC-05: Crop Disease Detection

Actor: Farmer, AMD Gateway

- Farmer uploads leaf photo
- Vision Brain analyses locally
- Disease/pest type returned
- Alert + treatment advice given

UC-06: Motor Guard

Actor: ESP32, AMD Gateway

- ESP32 sends PSI reading
- Acoustic Brain checks anomaly
- Emergency shutdown triggered
- All valves close instantly

UC-07: AI Farm Advisory

Actor: Farmer, Gemini API

- Farmer asks advisory question
- Gemini Bridge forwards query
- AI response returned
- Shown in dashboard chat

UC-08: Digital Ledger

Actor: Farmer, AMD Gateway

- Voice / action captured
- NLP extracts material + qty
- Stored in SQLite DB
- Dashboard shows history

UC-09: System Heartbeat

Actor: AMD Gateway

- Gateway publishes heartbeat/5s
- Dashboard shows ONLINE status
- Disconnection alert raised

Irrigation & Sensors · Farmer Manual Actions · Vision AI · Motor Safety · AI Advisory · Data & Logging